Technical Document #209: Retrieval Augmented Generation - Comprehensive Overvi

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Query decomposition breaks complex queries into simpler sub-queries. It improves retrieval quality for multi-faceted questions.

Vector databases store dense embeddings for semantic search. Pinecone, Weaviate, and ChromaDB are popular vector database solutions.

Hallucination detection identifies when LLMs generate factually incorrect information. It is critical for maintaining trust in RAG systems.

Chunking splits documents into smaller segments for embedding. Optimal chunk size balances context retention with retrieval precision.

Context window optimization selects the most relevant information to fit within the LLM's context limit. It maximizes information density.

Evaluation metrics for RAG include faithfulness, answer relevance, and context recall. They measure the quality of the retrieval-generation pipeline.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Key Takeaways:

- Vector databases store dense embeddings for semantic search.

- Hybrid search combines keyword-based and semantic search.

- Retrieval-Augmented Generation (RAG) combines information retrieval with text generation.